

Expressions

1. [March US 2023]

The ratio of a person's weight on Earth to the person's weight on the Moon is constant. An astronaut who weighs 540 newtons on Earth weighs 90 newtons on the Moon. Another astronaut weighs w newtons on Earth. Which of the following expressions represents this astronaut's weight, in newtons, on the Moon?

- A) $6w$ C) $\frac{6}{w}$
B) $0.6w$ D) $\frac{w}{6}$

2. [March US 2023]

Which of the following expressions is equivalent to $(\frac{a}{4} + \frac{b}{3})^2$?

- A) $\frac{a^2}{4} + \frac{b^2}{3}$
B) $\frac{a^2}{16} + \frac{b^2}{9}$
C) $\frac{a^2}{4} + \frac{ab}{12} + \frac{b^2}{3}$
D) $\frac{a^2}{16} + \frac{ab}{6} + \frac{b^2}{9}$

3. [March US 2023]

Jaqueline spent \$150 for supplies and gas to start a lawn-mowing service. She charges \$25 for each lawn she mows. In the first week Jaqueline made \$50 after the cost of supplies and gas was deducted. Which equation represents this situation, where x is the number of lawns Jaqueline mowed during the first week?

- A) $25x - 150 = 50$
B) $50 - 25x = 150$
C) $150 - 25x = 50$
D) $25x + 50 = 150$

4. [June 2023]

Which expression is equivalent to $(3x^3 - x^2 + 4)(5x^2 + 8x)$?

- A) $15x^5 + 19x^4 - 8x^3 - 20x + 32$
B) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32$
C) $15x^5 + 19x^4 - 8x^3 - 20x^2 + 32x$
D) $15x^5 + 29x^4 - 8x^3 - 20x^2 + 32x$

5. [June 2023]

If $\frac{6}{7}p + 12 = 54$ what is the value of $7p$?

6. [June 2023]

$$b^2 + 5c = 9d$$

The given equation relates the real numbers b , c , and d , where $d > \frac{5}{9}c$. Which equation correctly expresses b in terms of c and d ?

- A) $b = \frac{9d + 5c}{2}$
B) $b = \frac{9d - 5c}{2}$
C) $b = \pm\sqrt{9d + 5c}$
D) $b = \pm\sqrt{9d - 5c}$

7. [August US 2023]

If $x^3 + y^3 = 12$ and $x^3 - y^3 = 6$, what is the value of $x^6 - y^6$?

8. [August US 2023]

Which expression is equivalent to $2(xy^2 + x^2y) - 8(xy^2 - 4x^2y)$?

- A) $-6xy^2 + 34x^2y$
B) $-6xy^2 + 33x^2y$
C) $-6xy^2 - 30x^2y$
D) $-6xy^2 - 31x^2y$

9. [November 2023]

If $2x + 3 = 9$, what is the value of $4x - 4$?

10. [November 2023]

Which expression is equivalent to $(131y)^{\frac{1}{2}}$, where $y > 1$?

- A) $131\sqrt{y}$ C) $\sqrt{131y}$
B) $\sqrt{131}y$ D) $\sqrt{(131y)^2}$

11. [November 2023]

Which expression is equivalent to $(5a^2 - 3ab + 7b^2) - (4a^2 + ab - 2b^2)$

- A) $a^2 - 4ab + 9b^2$
B) $a^2 - 2ab + 5b^2$
C) $a^4 - 4a^2b^2 + 9b^4$
D) $a^4 - 2a^2b^2 + 5b^4$

12. [November 2023]

$$7x + 16 = \frac{4}{y}$$

The given equation relates the distinct positive numbers x and y . Which equation correctly expresses y in terms of x ?

- A) $y = 7x + 12$
B) $y = \frac{4}{7x + 16}$
C) $y = \frac{7x + 16}{4}$
D) $y = 4(7x + 16)$

13. [November 2023]

A total of 6 squares each have side length r . A total of 3 equilateral triangle each have side length t . None of these squares and triangles shares a side. The sum of the perimeters of all these squares and triangles is 75. Which equation represents this situation?

- A) $18r + 12t = 75$
B) $24r + 9t = 75$
C) $3r + 6t = 75$
D) $6r + 3t = 75$

14. [December 2023]

$$v = -\frac{w}{130x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

- A) $w = -130vx$
B) $w = -\frac{130v}{x}$
C) $w = -\frac{x}{130v}$
D) $w = v + 130x$

15. [December 2023]

Which expression is equivalent to $15(x + 10)$?

- A) $15x + 10$ C) $15x + 25$
B) $15x + 150$ D) $15x + 5$

16. [December 2023]

Which expression is equivalent to $60y + 90y$?

- A) $540y^2$ C) $150y^2$
B) $540y$ D) $150y$

17. [December 2023]

If $9(x + 3) = 81$, what is the value of $x + 3$?

- A) 6 C) 69
B) 9 D) 72

18. [March 2024]

Which expression is equivalent to

$$(9nr^2 + 3nr) + (6n^2r + 5nr)?$$

- A) $15n^2r^2 + 8nr$
B) $15n^3r^3 + 8n^2r^2$
C) $54n^3r^3 + 15n^2r^2$
D) $9nr^2 + 6n^2r + 8nr$

19. [March 2024]

$$v = -\frac{w}{161x}$$

The given equation relates the distinct positive numbers v , w , and x . Which equation correctly expresses w in terms of v and x ?

- A) $w = -161vx$
- B) $w = -\frac{161v}{x}$
- C) $w = -\frac{x}{161v}$
- D) $w = v + 161x$

20. [March 2024]

Which expression is equivalent to $16(x + 5)$?

- A) $16x + 5$
- B) $16x + 80$
- C) $16x + 21$
- D) $16x + 11$

21. [March 2024]

$$x = 2a(b + 5)$$

The given equation relates the positive numbers a , b , and x . Which equation correctly expresses a in terms of b and x ?

- A) $a = \frac{x}{2} - (b + 5)$
- B) $a = \frac{x}{2(b + 5)}$
- C) $a = \frac{2(b + 5)}{x}$
- D) $a = 2x(b + 5)$

22. [March 2024]

Which expression is equivalent to

$$\frac{y + 2}{x - 9} + \frac{y(x - 9)}{x^2y - 9xy}?$$

- A) $\frac{xy + y - 7}{x^3y - 18x^2y + 81xy}$
- B) $\frac{xy + 10y + 2}{x^2y - 9xy + x - 9}$
- C) $\frac{xy^2 + 3xy - 9y}{x^2y - 9xy}$
- D) $\frac{xy^2 + 3xy - 9y}{x^3y - 18x^2y + 81xy}$

23. [March 2024]

If $\frac{x}{y} = 63$ and $\frac{cx}{3y} = 63$, what is the value of c ?

24. [March 2024]

$$x = 2a(b + 3)$$

The given equation relates positive numbers x , a , and b . Which equation correctly expresses a in terms of b and x ?

- A) $a = \frac{x}{2} - 4b + 3$
- B) $a = \frac{x}{2(b + 3)}$
- C) $a = \frac{2(b + 3)}{x}$
- D) $a = 2x(b + 3)$

25. [March 2024]

If

$$8 + \frac{9(6 - x)}{4} = \frac{7(6 - x)}{3},$$

what is the value of $6 - x$?

26. [March 2024]

If

$$\frac{7}{8}p + 35 = 56,$$

what is the value of $8p$?

27. [March 2024]

The function f gives the product of a number, x , and a number that is 66 more than x . Which equation defines f ?

- A) $f(x) = x^2 + x + 66$
- B) $f(x) = x^2 + 66$
- C) $f(x) = x^2 + 66x$
- D) $f(x) = x^2 + 66x + 66$

28. [May 2024]

$$(5x^3 - 2) - (-6x^3 + 3)$$

The given expression is equivalent to $bx^3 - 5$ where b is a constant. What is the value of b ?

29. [May US 2024]

Which expression is equivalent to

$$(4x^2 + 6) - (3x^2)?$$

- A) 7
B) $7x^2 + 6$
C) $x^2 - 6$
D) $x^2 + 6$

30. [June 2024]

Which expression is equivalent

$$(x^3 + 6x^2 - 8x) + 8(x^2 + 4)?$$

- A) $x^3 + 4x^2 - 8x + 32$
B) $x^3 + 14x^2 - 8x + 32$
C) $x^3 + 8x^2 - 8x + 32$
D) $x^3 + 6x^2 - 8x + 32$

31. [June 2024]

The product of a positive number x and the number that is 1 less than x is equal to 342. What is the value of x ?

- A) 0.5
B) 19
C) 341
D) 343

32. [March 2024]

Which expression is equivalent to

$$\frac{y+4}{x-7} + \frac{y(x-7)}{x^2y-7xy}?$$

- A) $\frac{xy+y-3}{x^3y-14x^2y+49xy}$
B) $\frac{xy+8y+4}{x^2y-7x^2y+x-7}$
C) $\frac{xy^2+5xy-7y}{x^2y-7xy}$
D) $\frac{xy^2+5xy-7y}{x^3y-14x^2y+49xy}$

33. [June 2024]

If $\frac{x}{y} = 3$ and $\frac{5x}{ty} = 135$, what is the value of t ?

- A) $\frac{1}{9}$
B) $\frac{3}{5}$
C) $\frac{5}{3}$
D) 9

34. [June 2024]

A piece of string with a length of 35 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 5 more than 2 times the value of y . What is the value of x ?

35. [June 2024]

Which expression is equivalent to

$$(x^3 + 4x^2 - 3x) + 5(x^2 + 8)?$$

- A) $x^3 + 9x^2 - 3x + 40$
B) $x^3 + 4x^2 - 3x + 40$
C) $x^3 + 8x^2 - 3x + 40$
D) $x^3 + 5x^2 - 3x + 40$

36. [June 2024]

$$\frac{2}{c} - \frac{7}{d} = 5$$

The given equation relates the variables c and d , where c and d are greater than 1. Which equation correctly expresses d in terms of c ?

- A) $d = \frac{c}{7(2-5c)}$
B) $d = \frac{7c}{2-5c}$
C) $d = \frac{7(2-5c)}{c}$
D) $d = \frac{2-5c}{7c}$

37. [August 2024]

$$(9y + 9) - (5y + 7)$$

The given expression is equivalent to $by + c$, where b and c are constants. What is the value of b ?

38. [June 2024]

$$ak = \frac{b}{12}(7 + m)$$

The given equation relates the positive variables a, b, k , and m . Which equation correctly expresses m in terms of a, b , and k ?

A) $m = \frac{12(ak - 7)}{b}$

B) $m = ak - 7 - \frac{b}{12}$

C) $m = ak - \frac{7b}{12}$

D) $m = \frac{12ak}{b} - 7$

39. [August 2024]

If

$$a = 3k + 4r \text{ and } b = 9k - 12r + 6,$$

which expression is equivalent to $a - b$?

A) $-6k + 16r - 6$

B) $-6k + 16r + 6$

C) $-6k - 8r + 6$

D) $-6k - 8r - 6$

40. [August 2024]

The product of a positive number x and the number that is 5 more than x is 104. What is the value of x ?

A) 4

B) 8

C) 13

D) 26

41. [August 2024]

Which expression is equivalent to

$$15(x + 10)?$$

A) $15x + 10$

B) $15x + 150$

C) $15x + 25$

D) $15x + 5$

42. [August 2024]

$$\frac{20}{p} = \frac{20}{q} - \frac{20}{r} - \frac{20}{s}$$

The given equation relates the positive variables p, q, r , and s . Which of the following is equivalent to q ?

A) $p + r + s$

B) $20(p + r + s)$

C) $\frac{prs}{pr + rs + ps}$

D) $\frac{prs}{20p + 20r + 20s}$

43. [August US 2024]

Which expression is equivalent to

$$\frac{20\pi^2}{\pi^2 - x^2} - \frac{20\pi}{x + \pi}?$$

A) $\frac{20\pi}{\pi - x}$

B) $\frac{20\pi x}{\pi^2 - x^2}$

C) $\frac{20\pi^2 - 20\pi}{\pi^2 - x^2}$

D) $\frac{20\pi - 10}{(\pi - x^2)(x + 1)}$

44. [October 2024]

Which expression is equivalent to $4(x^2 + 6)$?

A) $4x^2 + 24$

B) $4x^2 + 10$

C) $4x^2 + 6$

D) $4x^2 - 2$

45. [August US 2024]

On average, a certain tree grows 51 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

- A) $\frac{17m}{4k}$
- B) $\frac{17k}{4m}$
- C) $\frac{612m}{k}$
- D) $\frac{612k}{m}$

46. [August US 2024]

Which expression is equivalent to

$$(6t^4 - 3t^2 + 3t - 6) - (2t^4 - 4t^3 + 6t^2 - 6)?$$

- A) $4t^4 + 4t^3 - 9t^2 - 3t - 12$
- B) $4t^4 + 4t^3 - 9t^2 - 3t$
- C) $4t^4 + t^3 - 9t - 12$
- D) $4t^4 + t^3 + 3t$

47. [August US 2024]

$$b - 22 = \frac{x}{y}$$

The given equation relates positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A) $x = \frac{by - 22}{y}$
- B) $x = by - 22$
- C) $x = by - 22y$
- D) $x = \frac{b - 22}{y}$

48. [October 2024]

The expression $72y^5 - 48y^4$ is equivalent to $ry^4(12y - 8)$, where r is a constant. What is the value of r ?

49. [October US 2024]

Which expression is equivalent to

$$11(x^2 - 100y^2)?$$

- A) $11(x - 2y)(x - 50y)$
- B) $11(x - 2y)(x + 50y)$
- C) $11(x - 10y)(x - 10y)$
- D) $11(x - 10y)(x + 10y)$

50. [October US 2024]

Which expression is equivalent to

$$(9c + 8) - (4c^3 - 6)?$$

- A) $4c^3 + 9c + 2$
- B) $4c^3 + 9c + 14$
- C) $-4c^3 + 9c + 2$
- D) $-4c^3 + 9c + 14$

51. [October US 2024]

The kinetic energy K , in joules, of an object is given by the formula $K = \frac{1}{2}mv^2$, where m is the mass of the object, in kilograms, and v is the velocity of the object, in meters per second. If the kinetic energy of a certain object can be found by using the formula $K = 32v^2$, what is the mass of the object, in kilograms?

- A) 1,024
- B) 64
- C) 32
- D) 16

52. [October US 2024]

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

53. [October 2024]

The total mass, in kilograms, of r identical objects is t . Which expression represents the total mass, in kilograms, of $146r$ of these objects?

- A) $146 - t$ C) $\frac{t}{146}$
 B) $146 + t$ D) $146t$

54. [October US 2024]

Which expression is equivalent to

$$11x^3 + 2x^2 + 7x?$$

- A) $20(x^3 + x^2 + x)$
 B) $x(11x^2 + 2x + 7)$
 C) $7x(11x^3 + 2x^2 + 1)$
 D) $20(x^3 + x^2 + x)$

55. [October US 2024]

$$\frac{49}{p} = \frac{49}{q} - \frac{49}{r} - \frac{49}{s}$$

The given equation relates the positive variables p , q , r and s . Which of the following is equivalent to q ?

- A) $p + r + s$
 B) $(49p + r + s)$
 C) $\frac{prs}{pr + ps + rs}$
 D) $\frac{prs}{49p + 49r + 49s}$

56. [October US 2024]

If

$$\frac{j}{k} = 62 \text{ and } \frac{96j}{12xk} = 62,$$

what is the value of x ?

- A) $\frac{1}{8}$ C) 8
 B) $\frac{6}{31}$ D) 96

57. [October 2024]

If $\frac{42}{x} = 7x$, what is the value of $7x^2$?

- A) 6 C) 42
 B) 7 D) 294

58. [October 2024]

$$b - 72 = \frac{x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A) $x = \frac{b - 72}{y}$
 B) $x = by - 72$
 C) $x = \frac{by - 72}{y}$
 D) $x = by - 72y$

59. [October US 2024]

Which expression is equivalent to $63x^2$?

- A) $63(1 + x^2)$ C) $(7x^2)(9x^2)$
 B) $(7x)(9x)$ D) $(63x^2)(x)$

60. [October US 2024]

Which expression is equivalent to

$$(3nr^2 + 4nr) + (8n^2r + 2nr)?$$

- A) $11n^2r^2 + 6nr$
 B) $11n^3r^3 + 6n^2r^2$
 C) $24n^3r^3 + 8n^2r^2$
 D) $3nr^2 + 8n^2r + 6nr$

61. [November 2024]

$$v = -\frac{w}{188x}$$

The given equation relates the distinct positive numbers v , w and x . Which equation correctly expresses w in terms of v and x ?

A) $w = -188vx$

C) $w = -\frac{x}{188v}$

B) $w = -\frac{188v}{x}$

D) $w = v + 188x$

62. [November 2024]

One gallon of sealant costs \$23 and will cover 250 square feet of a surface. A deck has a total surface area of d square feet. Which of the following represents the cost c , in dollars, of the sealant needed to cover the deck twice?

A) $c = \frac{250d}{23}$

C) $c = 23\left(\frac{d}{125}\right)$

B) $c = \frac{500d}{23}$

D) $c = 23\left(\frac{d}{250}\right)$

63. [December 2024]

Which expression is equivalent to

$$158t^3 - 24t^2u?$$

A) $2t(79t^2 - 12u)$

B) $2t^2(79t - 12u)$

C) $2tu(79t^2 - 12)$

D) $2t^2u(79t - 12)$

64. [December 2024]

$$PC = N(44 - C)$$

The given equation relates the variables P , N , and C . Which expression represents the value of C for distinct positive values of P and N ?

A) $\frac{C(N + P)}{N}$

B) $\frac{PC}{44 - C}$

C) $\frac{44N}{N + P}$

D) $\frac{44N}{P + 1}$

65. [December 2024]

$$49 = \frac{2K}{x^2}$$

For an object with a mass of 49 kilograms, the given equation relates the kinetic energy K , in joules, of the object to the object's speed v , in meters per second, where K and v are positive. Which equation correctly expresses this object's speed, in meters per second, in terms of the object's kinetic energy, in joules?

A) $v = \sqrt{\frac{2K}{49}}$

C) $v = \frac{49^2}{2K}$

B) $v = \sqrt{\frac{49}{2K}}$

D) $v = \frac{2K}{49^2}$

66. [December 2024]

On average, a certain tree grows 87 centimeters every m months. At this rate, which expression represents the number of centimeters, on average, the tree grows every k years?

A) $\frac{29m}{4k}$

C) $\frac{1,044m}{k}$

B) $\frac{29k}{4m}$

D) $\frac{1,044k}{m}$